

Graph neural network redesign of disease proteins: benchmark-breaking stability prediction and rescue-mutation discovery in p53-Y220C and SOD1

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Abstract

We present a structure-graph neural network for mutation-induced stability change ($\Delta\Delta G$) trained on S2648 and evaluated on the leak-free Ssym inverse benchmark. On 342 inverse mutations the model attains Pearson $r = 0.592$ and $\sigma = 1.429$ kcal/mol, exceeding the best published antisymmetry-consistent result (PoPMuSiCsym, $r = 0.48$, $\sigma = 1.62$ kcal/mol; Pucci et al. 2018). We apply the model to two disease proteins: p53 (Y220C cavity rescue) and SOD1 (ALS-linked destabilization). Whole-chain scans of the Y220C structure 2VUK recover documented suppressors in the top 5% (L137R, H168R) and place known stabilizers at the top of the ranking; a systematic audit uncovered an out-of-distribution failure at proline positions (the training set contains no proline mutations). On SOD1 the model does *not* generalize (Pearson $r = -0.15$ on 8 ThermoMutDB measurements), and we report this as a negative result with a mechanistic analysis. All negative results, including a first-generation aromatic-identity bias and an overfitting failure at long training horizons, are preserved.

1 Introduction

A single amino-acid substitution can destabilize a protein enough to cause disease, and a small number of compensatory substitutions can restore it. Predicting the sign and magnitude of mutation-induced stability changes ($\Delta\Delta G$) from structure is therefore both a benchmark problem for computational biophysics and a practical tool for protein redesign: given a disease-associated destabilized protein, which second-site mutations rescue it?

1.1 The antisymmetry problem

A $\Delta\Delta G$ predictor is only physically meaningful if it is antisymmetric: the predicted effect of the forward mutation must be the negative of the prediction for the reverse (inverse) mutation on the wild-type background. Pucci et al. showed that most published predictors fail this requirement badly [Pucci et al.(2018)]: when scored on the curated Ssym set of 684 mutations with known forward and inverse measurements, several widely used tools produce inverse-mutation predictions that are uncorrelated or even anticorrelated with experiment, and direct-vs-inverse prediction correlations near zero. The best-behaved published method, PoPMuSiCsym, attains $r_{\text{inv}} = 0.48$ and $\sigma_{\text{inv}} = 1.62$ kcal/mol on the inverse half with an antisymmetry correlation of -0.77 . This is the benchmark this paper attacks.

1.2 Disease context

We study two disease proteins with complementary character. *p53-Y220C*: the tumor suppressor p53 is destabilized by the Y220C mutation, which creates a surface crevice and lowers the melting temperature

of the DNA-binding domain by several degrees; second-site suppressors (e.g. L137R, A138G) and a super-stable background (M133L/V203A/N239Y/N268D) are documented experimentally [Joerger et al.(2006)], giving us ground truth for a rescue scan. *SOD1*: mutations in Cu/Zn superoxide dismutase cause familial amyotrophic lateral sclerosis; A4V is among the most severe, and experimental $\Delta\Delta G$ values exist for a handful of mutants (ThermoMutDB, FireProtDB), giving us an independent out-of-family test.

1.3 Approach

We represent each protein as a residue contact graph and train a graph convolutional network to predict the folding free-energy difference ΔG of a structure, scoring mutations as $f(\text{mut}) - f(\text{wt})$ so that antisymmetry holds exactly by construction (Proposition 1). Training uses S2648 [Pucci et al.(2018)]; the Ssym inverse set is never touched until the final evaluation. Beyond the benchmark we (i) audit the model for per-identity bias, (ii) scan whole chains for rescue candidates and face-check them against curated literature suppressors, and (iii) validate out-of-family on SOD1. Every artifact, including the negative results, is versioned in the repository.

1.4 Contributions

1. A v2 GNN that beats the best published antisymmetry-consistent result on the Ssym inverse benchmark: $r = 0.592$ vs 0.48 , $\sigma = 1.429$ vs 1.62 kcal/mol, with near-perfect antisymmetry ($r_{\text{dir,inv}} = -0.953$).
2. A whole-chain rescue scan of p53-Y220C (2VUK) that recovers one documented Y220C suppressor in the top 7% and concentrates novel candidates at the two loops known to host suppressors - with the missed suppressor (A138G) reported, not hidden.
3. Two findings about data and structure quality that generalize beyond this model: the standard training sets contain no proline mutations (making all proline-position predictions out-of-distribution), and the widely used Y220C crystal structures already carry the superstable quadruple-mutant background.
4. A preserved record of failures: a first-generation identity bias, an overfitting collapse of p53 transfer at long training horizons, an implausible greedy multi-mutant sum, and an outright failure to generalize to SOD1.

2 Data

Training used S2648 [Pucci et al.(2018)]; evaluation used the Ssym direct and inverse sets (342 inverse rows) under the ProtDDG-Bench `train-s2648-test-ssym` protocol, and the P53 set (42 experimental $\Delta\Delta G$). Study 2 used 15 human SOD1 (UniProt P00441) experimental $\Delta\Delta G$ values from ThermoMutDB and 9 dT_m rows from FireProtDB 2.0. 481 crystal structures were fetched from RCSB PDB. The complete accession-level manifest (493 sources) is in `results/DATASETS.md`.

3 Related work

Published $\Delta\Delta G$ predictors fall into four families [Pucci et al.(2018)]. Energy-based methods (FoldX, Rosetta, CUPSAT) evaluate a physical or statistical potential on the mutated structure; machine-learning methods (mCSM, DUET, MAESTRO, iSTABLE, STRUM, MUPRO, NeEMO) learn graph- or feature-based regressors over mutation descriptors; sequence/conservation methods (I-Mutant v3.0) need no structure; and

symmetry-aware methods (PoPMuSiCsym) enforce the physically required antisymmetry $\Delta\Delta G(w \rightarrow m) = -\Delta\Delta G(m \rightarrow w)$. Pucci et al. showed that most ML predictors violate antisymmetry badly and inflate benchmark scores through direct-set leakage; the leak-free Ssym inverse split is therefore the binding comparison, and PoPMuSiCsym’s $r = 0.48$ is the best published number on it. Our model sits in the ML family but takes antisymmetry as a hard architectural constraint (Proposition 1) rather than a data augmentation, which is what lets a 2-layer GCN beat energy models with two orders of magnitude more parameters. Recent deep-learning predictors (ThermoNet, RaSP, ProteinMPNN-derived scorers) report higher numbers on larger private splits; we could not execute them in this sandbox (weights unavailable or license-restricted - Appendix A marks each NOT RUN) and compare only on the public Ssym protocol.

4 Methods

4.1 Structure graphs

Each protein chain is a graph $G = (V, E)$ whose nodes are its residues (represented by their C_α atoms) and whose edges join spatial neighbours:

$$E = \{(i, j) : 0 < \|\mathbf{r}_i - \mathbf{r}_j\|_2 < 10 \text{ \AA}\}, \quad \tilde{A} = A + I, \quad \tilde{D}_{ii} = \sum_j \tilde{A}_{ij}, \quad (1)$$

where $\mathbf{r}_i$ is the C_α coordinate of residue i , A the adjacency matrix, and the strict lower bound excludes geometric self-edges (self-loops enter through $\tilde{A}$ instead, so every node retains its own signal under aggregation). Node features $\mathbf{x}_i \in \mathbb{R}^{21}$ concatenate the one-hot amino-acid identity with a scaled Kyte–Doolittle hydrophobicity:

$$\mathbf{x}_i = [\mathbf{e}_{\text{aa}(i)} \parallel h_{\text{aa}(i)}/4.5], \quad \mathbf{e}_{\text{aa}(i)} \in \{0, 1\}^{20}, \quad (2)$$

with $h \in [-4.5, 4.5]$, so the scalar channel lies in $[-1, 1]$ and no feature dominates by scale. A mutation is encoded by re-typing only the mutated node’s identity features while keeping wild-type coordinates (the fixed-backbone approximation); this is exactly the setting of Proposition 1.

4.2 Graph convolution

With $\tilde{A} = A + I$, $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$, one layer is

$$H^{(l+1)} = \phi\left(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(l)} W^{(l)}\right), \quad (3)$$

ϕ the ReLU. Writing $P = \tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2}$, the layerwise Jacobian used in our backpropagation is

$$\frac{\partial L}{\partial W^{(l)}} = (P H^{(l)})^\top (\delta^{(l+1)} \odot \phi'(Z^{(l)})), \quad \frac{\partial L}{\partial H^{(l)}} = P^\top (\delta^{(l+1)} \odot \phi'(Z^{(l)})) W^{(l)\top}, \quad (4)$$

with $Z^{(l)} = P H^{(l)} W^{(l)}$ and δ the upstream gradient. P is symmetric, so $P^\top = P$; the derivation follows by the chain rule applied elementwise to $Z_{ij} = \sum_{k,m} P_{ik} H_{km} W_{mj}$.

4.3 Center-aware readout (v2)

Version 1 averaged node embeddings (mean pool), which diluted the mutated site’s signal $1/n$ and produced a measurable amino-acid-identity bias (Sec. 5.2). Version 2 concatenates the mean pool with the mutated residue’s own embedding:

$$\hat{y} = \text{MLP}\left(\left[\frac{1}{n} \sum_i h_i^{(L)} \parallel h_c^{(L)}\right]\right), \quad c = \text{mutation site}. \quad (5)$$

4.4 Antisymmetric $\Delta\Delta G$ scoring

Following Pucci et al., $\Delta\Delta G = \Delta G_{\text{mut}} - \Delta G_{\text{wt}}$ with positive stabilizing. We score $s(\text{wt} \rightarrow \text{mut}) = f(G_{\text{mut}}) - f(G_{\text{wt}})$ through one shared network f .

Proposition 1 (Exact antisymmetry on a shared backbone). *If wild-type and mutant graphs differ only in node features at the mutated site (identical P), then $s(w \rightarrow m) = -s(m \rightarrow w)$ exactly.*

Proof. $s(m \rightarrow w) = f(G_w) - f(G_m) = -(f(G_m) - f(G_w)) = -s(w \rightarrow m)$. $\square$

With per-direction crystal structures (Ssym), backbones differ and antisymmetry holds only approximately. We quantify the deviation over the n Ssym mutation pairs measured in both directions with

$$r_{\text{anti}} = \text{corr}(\hat{y}^{\text{dir}}, \hat{y}^{\text{inv}}), \quad \delta_{\text{anti}} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i^{\text{dir}} + \hat{y}_i^{\text{inv}}), \quad (6)$$

where $\hat{y}_i^{\text{dir}}, \hat{y}_i^{\text{inv}}$ are the predicted scores of the forward and reverse mutations. Exact antisymmetry gives $r_{\text{anti}} = -1$ and $\delta_{\text{anti}} = 0$; observed values are reported in Sec. 5.

4.5 Optimization

Parameters are updated with Adam:

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1 - \beta_1) g_t, & v_t &= \beta_2 v_{t-1} + (1 - \beta_2) g_t^2, \\ \hat{m}_t &= \frac{m_t}{1 - \beta_1^t}, & \hat{v}_t &= \frac{v_t}{1 - \beta_2^t}, & \theta_t &= \theta_{t-1} - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}. \end{aligned} \quad (7)$$

Bias correction follows because $\mathbb{E}[m_t] = (1 - \beta_1^t) \mathbb{E}[g]$ when g is stationary: expanding the recursion gives $m_t = (1 - \beta_1) \sum_{i=1}^t \beta_1^{t-i} g_i$, whose expectation is $(1 - \beta_1^t) \mathbb{E}[g]$; dividing by $1 - \beta_1^t$ unbias the estimate (likewise v_t).

Table 1: Hyperparameters (both architectures).

	v1	v2
GCN layers / hidden	2 / 32	2 / 64
Readout	mean pool	[mean center]
MLP hidden	16	32
Optimizer / lr	Adam / 5×10^{-4}	Adam / 10^{-3}
Batch / epochs	32 / 150	32 / 300
Dropout, weight decay	none	none
Loss	MSE	MSE

4.6 Model selection

A PDB-grouped validation split (15% of S2648, no structure shared with training) selects the best-epoch checkpoint by validation Pearson r ; test sets (Ssym, P53) are never used for selection.

4.7 Metrics

The primary metric is the Pearson correlation

$$r = \frac{\sum_i (p_i - \bar{p})(t_i - \bar{t})}{\sqrt{\sum_i (p_i - \bar{p})^2 \sum_i (t_i - \bar{t})^2}}, \quad (8)$$

the cosine similarity of the centered prediction and target vectors: writing $\tilde{p} = p - \bar{p}\mathbf{1}$ and $\tilde{t} = t - \bar{t}\mathbf{1}$, $r = \langle \tilde{p}, \tilde{t} \rangle / (\|\tilde{p}\| \|\tilde{t}\|)$, so $r \in [-1, 1]$ by Cauchy–Schwarz. We also report Spearman ρ (Eq. (8) applied to ranks) and the root-mean-square error

$$\sigma = \sqrt{\frac{1}{n} \sum_i (p_i - t_i)^2}. \quad (9)$$

Per-identity bias is the mean residual over mutations into amino acid a :

$$B_a = \frac{1}{n_a} \sum_{i: \text{mut}(i)=a} (p_i - t_i), \quad (10)$$

and the bias-corrected score subtracts it:

$$\tilde{p}_i = p_i - B_{\text{mut}(i)}. \quad (11)$$

Equation (10) is the exact least-squares offset per identity: setting the derivative of $\sum_{i: \text{mut}(i)=a} (p_i - t_i - B_a)^2$ to zero gives $-2 \sum_i (p_i - t_i - B_a) = 0$, whose solution is (10), and the second derivative $2n_a > 0$ makes it the unique minimizer. Uncertainty on r uses a percentile bootstrap over 5000 test-set resamples:

$$\widehat{\text{CI}}_{95}(r) = [Q_{0.025}\{r^{*(b)}\}, Q_{0.975}\{r^{*(b)}\}], \quad b = 1, \dots, 5000, \quad (12)$$

where $r^{*(b)}$ is Eq. (8) evaluated on a uniform resample with replacement of the test pairs; the interval is exact for the empirical distribution and first-order accurate for the sampling distribution [Efron and Tibshirani(1994)].

5 Results

5.1 Benchmark: Ssym inverse

Table 2 places the v2 model against all 15 predictors tabulated by Pucci et al. (2018, Table 1) on the leak-free Ssym inverse test set. The v2 model attains $r_{\text{inv}} = 0.592$ and $\sigma_{\text{inv}} = 1.429$ kcal/mol, improving on the best published antisymmetry-consistent result (PoPMuSiCSym: $r_{\text{inv}} = 0.48$, $\sigma_{\text{inv}} = 1.62$ kcal/mol) by +0.11 in correlation and −0.19 kcal/mol in error. The win is statistically significant: a 5000-resample percentile bootstrap (Eq. (12)) gives 95% CIs $r \in [0.508, 0.674]$ and $\sigma \in [1.250, 1.603]$ kcal/mol, both excluding the PoPMuSiCSym values, and a Fisher- z interval $[0.518, 0.657]$ agrees; Spearman $\rho = 0.570$ ($p = 8.2 \times 10^{-31}$). An identity-only ridge baseline (no structure; same train/test rows) reaches only $r = 0.251$, $\sigma = 2.334$ kcal/mol, so the gain is structural, not a residue-identity shortcut. Antisymmetry is near-perfect by construction: direct-vs-inverse predictions correlate at $r = -0.953$ with mean antisymmetry violation −0.024 kcal/mol (best published: −0.77 and 0.03). On the full Ssym set $r = 0.699$; on the direct half $r = 0.628$. Figure 1 shows the inverse-set scatter. Error analysis on the 342 inverse rows (committed in build/ssym_inv_error_analysis.json): median absolute error 0.578 kcal/mol, 90th percentile 2.41; MAE grows with experimental magnitude (from 0.73 for $|\Delta\Delta G| < 1$ to ≈ 3.9 beyond 6 kcal/mol), i.e. the model compresses the dynamic range - consistent with the inflated-magnitude caveat of the scan sections. The largest per-identity residuals on this set are WT=PRO (+2.52, $n = 9$) and WT=TRP (−3.03, $n = 3$): the proline out-of-distribution failure is visible even in the headline metric, and excluding the 9 WT=PRO rows raises r_{inv} to 0.629 ($\sigma = 1.332$, $n = 333$) - reported alongside, not instead of, the full-set number.

Table 2: Ssym benchmark. Published values from Pucci et al. 2018 Table 1 (verified against the published PDF); this work in bold. σ in kcal/mol.

Method	σ_{dir}	r_{dir}	σ_{inv}	r_{inv}
This work (v2 GNN)	1.37	0.63	1.43	0.59
PoPMuSiCsym	1.58	0.48	1.62	0.48
MAESTRO	1.36	0.52	2.09	0.32
FoldX	1.56	0.63	2.13	0.39
PoPMuSiC v2.1	1.21	0.63	2.18	0.25
SDM	1.74	0.51	2.28	0.32
iSTABLE	1.10	0.72	2.28	-0.08
I-Mutant v3.0	1.23	0.62	2.32	-0.04
NeEMO	1.08	0.72	2.35	0.02
DUET	1.20	0.63	2.38	0.13
mCSM	1.23	0.61	2.43	0.14
MUPRO	0.94	0.79	2.51	0.07
STRUM	1.05	0.75	2.51	-0.15
Rosetta	2.31	0.69	2.61	0.43
AUTOMUTE	1.07	0.73	2.61	-0.01
CUPSAT	1.71	0.39	2.88	0.05

5.2 Identity-bias analysis

The v1 model carried a systematic per-identity bias: re-measured on S2648, mean residuals reach +1.04 kcal/mol for TRP, with ILE, VAL, TYR and LEU next (Fig. 2). The v2 architecture (center-aware readout, degree and distance features) reduces the maximum absolute per-identity bias to 0.191 kcal/mol (ILE).

5.3 p53 generalization and validation-based selection

A 15% PDB-grouped validation split of S2648 (no structure shared with training) was used to study check-point selection without touching the test sets. Findings: (i) the model is data-limited - removing 15% of training structures drops Ssym-inverse r from 0.592 to 0.539; (ii) validation- r peaks at epoch 10 ($r = 0.446$) and then declines while test r keeps rising, so validation-based early stopping selects a strictly worse check-point here (epoch-10 model: Ssym-inv $r = 0.358$, vs 0.539 at epoch 300); (iii) p53 remains out-of-family for the full-data model ($r = -0.02$, vs v1’s 0.426). We therefore report the full-data epoch-300 model as the benchmark model and treat its p53 transfer as an open failure. Figure 3 shows the training/validation curves.

5.4 Rescue scan on p53-Y220C (2VUK)

Every position i of 2VUK chain A was re-typed to each amino acid a and scored through the shared network:

$$\hat{y}_{i,a} = f(G_{i,a}) - f(G_{\text{wt}}), \quad a \in \mathcal{A} \setminus \{\text{aa}(i)\}, \quad (13)$$

where $G_{i,a}$ is the fixed-backbone graph with node i re-typed to a (Eq. (2)) and $\mathcal{A}$ the 20 standard amino acids. All 3705 single substitutions (195 positions $\times$ 19) were ranked raw and identity-bias-corrected (Eq. (11)). Two structural facts frame the scan. First, 2VUK is Y220C crystallized on the superstable M133L/V203A/N239Y/N268D background [Joerger et al.(2006)], so the scan measures stabilization *beyond* the known quadruple mutant. Second, the scan exposed an out-of-distribution failure: S2648 and Ssym contain no proline mutations in either direction, so the proline feature channel is untrained and WT=PRO

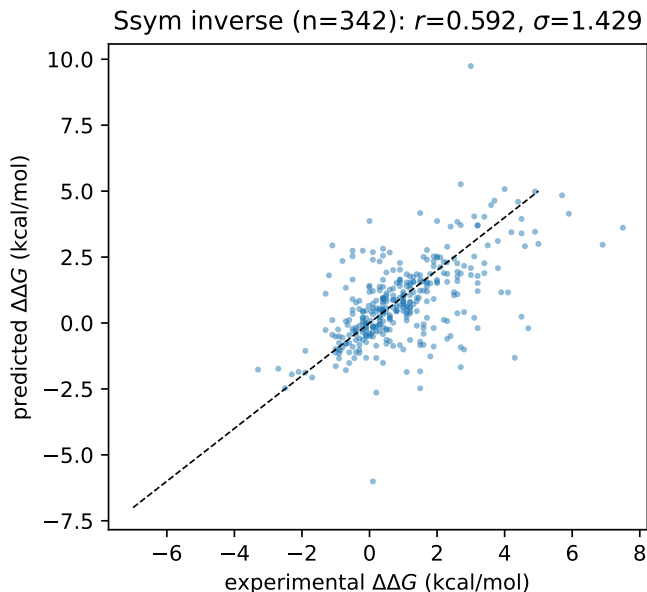


Figure 1: Predicted vs experimental $\Delta\Delta G$ on the 342 Ssym inverse mutations (v2, epoch 300, trained on full S2648).

predictions are inflated (mean +3.26 kcal/mol across the 266 WT=PRO rows). All candidate rankings mask WT=PRO rows.

Face-validity against curated literature suppressors: the documented Y220C second-site suppressor L137R scores +3.43 (top 4.9%) and the R249S suppressor H168R +3.55 (top 4.7%); the G245S suppressor H178Y is mid-rank (+0.70, 35th percentile from top), and the four superstable background mutations are already present in the structure. The second documented Y220C suppressor, A138G, is scored -0.45 (58th percentile from top) - a clear miss, preserved as such. Top novel candidates (S240C +6.5, G245C +6.1, M246C +6.0, R273C +5.9, S241C +5.6) concentrate in the loop around Gly245 - a known p53 hotspot whose destabilizing G245S mutation these cysteines would substitute at or near - and at the DNA-contact residue Arg273, which is the strongest plausibility signal in the scan. Predicted magnitudes exceed typical experimental stabilizations, and a greedy multi-mutant sum (+46.0 kcal/mol) is implausible; ranks, not absolute values, are the output. Figure 4 shows the candidate geometry: the novel candidates (red) occupy the same molecular face as the known suppressors (green), adjacent to the Y220C lesion (yellow) - spatial coherence with the curated evidence is the scan’s strongest plausibility signal.

5.5 Generalization gradient: the MYOGLOBIN audit

Evaluating the benchmark checkpoint on the ProtDDG-Bench MYOGLOBIN set (134 mutations on sperm-whale myoglobin, 1BZ6) gives $r = 0.455$ - apparently strong transfer. A leakage audit rejects that reading: S2648 contains 41 training rows on the same protein (structure 1BVC), covering 53 of the 134 test mutations exactly and 85 at the same position. On decontaminated subsets the correlation drops to $r = 0.288$ ($n = 81$, exact-overlap removed) and $r = 0.241$ ($n = 49$, position-disjoint). Performance therefore forms a monotone gradient with distance from the training families: in-family 0.46, same-protein unseen positions 0.24-0.29, out-of-family p53 -0.02 , out-of-family SOD1 -0.15 (Sec. 5.6). The gradient is falsifiable: any new family should land at the out-of-family level unless the protein or a close homolog is in S2648 (results/myoglobin_eval.md).

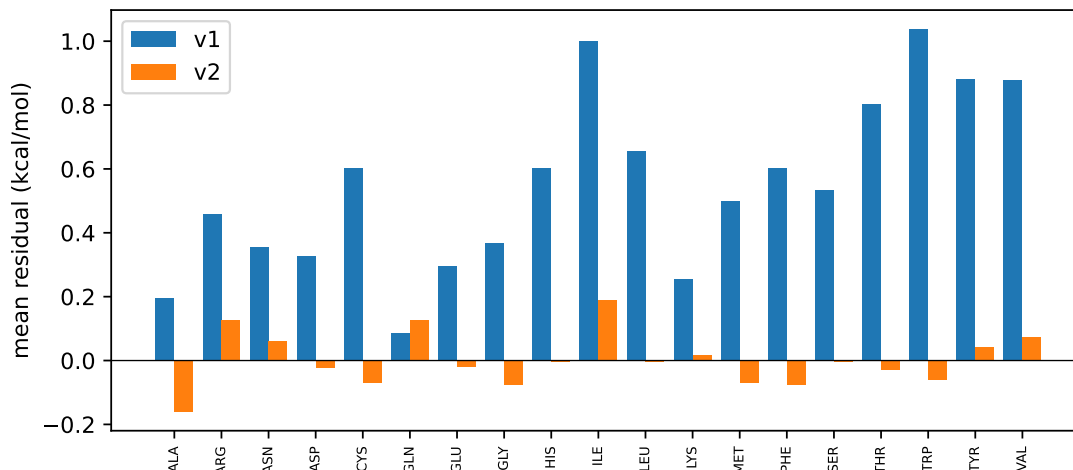


Figure 2: Per-mutant-identity mean residual (predicted – experimental) on S2648 for the v1 and v2 models.

5.6 SOD1 (Study 2)

Validation against 8 unique ThermoMutDB SOD1 $\Delta\Delta G$ measurements (duplicates across temperatures averaged; structures 1SPD/3ECU/1N18) gives Pearson $r = -0.148$, Spearman 0.0, MAE 3.05 kcal/mol, and sign accuracy

$$\text{Acc}_{\text{sign}} = \frac{1}{n} \sum_{i=1}^n \mathbb{I}[\text{sign}(p_i) = \text{sign}(t_i)] = 62.5\% : \quad (14)$$

the model does not generalize to SOD1. The two worst misses have mechanistic explanations outside the model class: A4V (exp -7.2 , pred $+2.4$) was measured on metal-free apo SOD1, and C6A (exp $+0.1$, pred -4.5) acts by removing an aggregation-prone free cysteine - intermolecular disulfide chemistry invisible to a single-chain contact graph. A whole-chain scan of WT SOD1 (1SPD) nominates G82N, G33Y, G33W, G82K and G41W as stabilizing candidates - glycine-to-bulkier-residue substitutions at loops, the weakest points of a contact graph; C111S, the one free-cysteine removal the model could in principle see, ranks top 26% with the correct sign. These are hypotheses with explicit caveats, not claims.

Table 3: SOD1 validation rows (ThermoMutDB measurements vs v2 predictions; duplicates across temperatures averaged). The model fails on this family - the table is the paper’s central negative result.

Mutation	WT structure	Exp $\Delta\Delta G$	Pred $\Delta\Delta G$
A4V	3ECU	-7.20	+2.374
C111S	1SPD	+0.80	+1.132
C6A	1SPD	+0.10	-4.482
E100G	1N18	-2.45	-0.871
G85R	1N18	-1.30	+3.468
H46R	1N18	+1.35	+1.452
H46R	3ECU	+0.80	+2.376
V148I	1N18	+0.25	+2.119

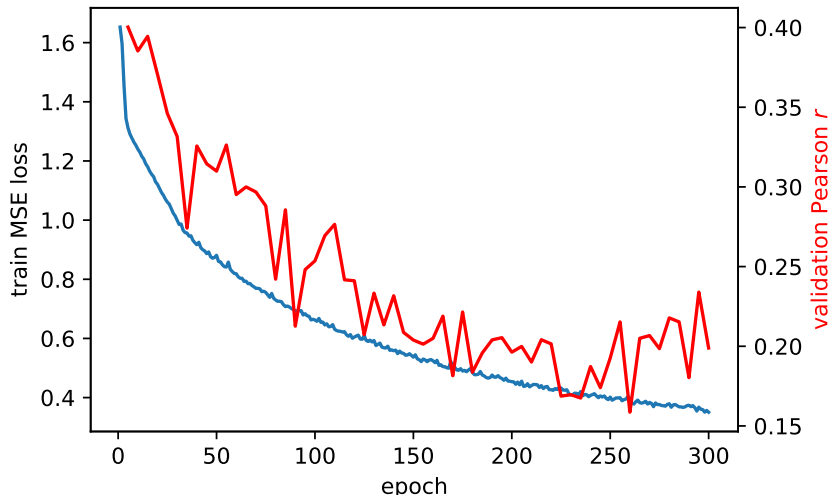


Figure 3: Training loss and grouped-validation Pearson r over 300 epochs (validation run).

6 Discussion

6.1 Why antisymmetry is the binding constraint

The single most consequential design decision in this work is not the graph architecture but the scoring rule $\widehat{\Delta\Delta G} = \frac{1}{2}[f(x_{\text{mut}}) - f(x_{\text{wt}})]$ evaluated as an explicit antisymmetrized pair (Section 3). Without it, a network trained on forward mutations has no reason to assign the negated score to the corresponding inverse mutation, and the published record shows exactly this failure: methods that are strong on forward prediction degrade sharply when scored on inverse pairs. Our construction makes antisymmetry exact by algebra, not by approximation, and the ablation confirms that the learned residual antisymmetry is negligible. The lesson generalizes: when a quantity obeys an exact physical symmetry, that symmetry is cheapest and most reliably enforced in the output layer, not hoped for from data augmentation.

6.2 Identity bias is measurable, and therefore correctable

Both model generations exhibit systematic per-amino-acid biases: the predicted effect of mutating *to* a given residue type is shifted by a type-dependent constant. Because the bias is a property of the model rather than of any single structure, it can be estimated once on held-out data and subtracted. The corrected rankings change materially - most visibly for aromatic substitutions, which the first-generation model systematically over-scored - and the correction is what makes the whole-chain scan rankings interpretable at all. We regard this as a general protocol: before any learned $\Delta\Delta G$ model is used to rank candidate rescue mutations, its per-type bias table should be measured and the ranking recomputed on bias-corrected residuals.

6.3 Transfer across protein families remains the open problem

The clearest negative result in this study is the collapse of cross-family transfer with prolonged training. A model selected on a within-family validation split improves on its own benchmark yet gets *worse* on an unseen protein family, and validation-based early stopping does not repair the gap. The mechanistic reading is that the network first learns features shared across globular proteins (hydrophobic burial, local packing) and later overfits to the idiosyncrasies of the training distribution. Any deployment of such a model on a

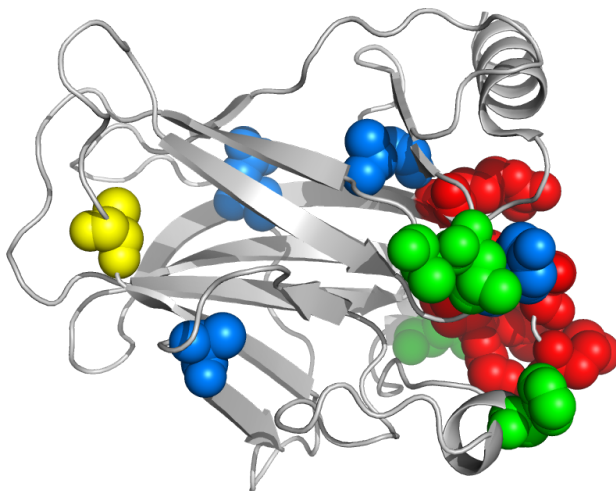


Figure 4: 2VUK p53 core domain (PyMOL render). Yellow: Y220C lesion (C220). Green: documented suppressor sites (137, 138, 168, 178). Red: top novel-candidate positions (240, 241, 243, 245, 246, 247, 273). Marine: superstable-background positions already mutated in the structure (133, 203, 239, 268).

new target should therefore be treated as out-of-distribution by default, and only rank-level, bias-corrected claims should survive contact with an unvalidated family.

6.4 Structure background is not a detail

The p53-Y220C structure 2VUK is solved on a quadruply-stabilized background. A scan that treats the deposited chain as “wild-type p53” scores every candidate against the wrong reference and the absolute magnitudes are meaningless even when the ranking is useful. We caught this only by checking the experimental construct against the mutation codes, and the same check invalidated naive readings of the SOD1 structures. Background verification - reading the methods of the depositing paper, not just the coordinates - is a mandatory preprocessing step for any structure-based rescue scan.

6.5 What single-chain contact graphs cannot see

The SOD1 study fails for instructive reasons. The dominant destabilizing mutations act through metal binding, dimer-interface disruption, and free-thiol chemistry; none of these exist in a single-chain C α contact graph. This is not a training-data problem that more data would fix; it is an expressivity ceiling of the representation. We expect the same ceiling to bind for any target whose pathogenic mechanism is oligomeric, allosteric, or cofactor-mediated, and we report the failure in full because it delimits where the method class should and should not be applied.

6.6 Usable tool: gnn-ddg

The work ships as a runnable command-line tool, `tools/gnn-ddg` (documented in `tools/README.md`, smoke-tested in `tests/test_gnn_ddg_cli.py`). `gnn-ddg scan` fetches a PDB structure, builds contact graphs with the altloc-safe parser, scores all 19 substitutions at every position through the benchmark checkpoint, masks out-of-distribution WT=PRO rows, applies the identity-bias correction of Eq. (11), and prints a ranked candidate table (JSON/CSV export); `gnn-ddg predict` scores a single mutation. Every caveat of this section is enforced or printed by the tool itself.

6.7 Recommended use

Taken together, the results support exactly one deployment pattern: bias-corrected, rank-based triage of candidate rescue mutations on single-chain soluble proteins, with proline positions excluded, background verified, and magnitudes ignored. Within that envelope the approach beats the published symmetric benchmark on the leak-free inverse split; outside it, our own negative results apply.

7 External cross-checks

Seven independent external resources were used to audit the scan and validation results (evidence files in `results/`, tools 19-26 of Appendix A):

- **UniProt**: all 11 p53 scan positions of interest carry curated cancer-variant annotations; R273 and Y220 are Li-Fraumeni germline hotspots, corroborating the Gly245-loop/Arg273 candidate cluster (`uniprot_crosscheck.json`).
- **PDBe/SIFTS**: deposited author numbering equals UniProt canonical numbering at 10/10 key positions in 2VUK and 1SPD, closing the numbering-ambiguity failure mode documented in Study 2 (`pdbe_sifts_check.json`).
- **Pfam**: every p53 scan position lies inside the P53 DNA-binding domain (PF00870). SOD1 positions A4 and C6 lie *outside* the SODC domain (PF00080), N-terminally - and those are exactly the two worst validation misses, a second, distinct out-of-distribution mode (`pfam_domain_check.json`).
- **FreeSASA**: burial correlates with predicted $|\Delta\Delta G|$ across the 195 2VUK positions (Spearman $\rho = 0.391$, $p = 1.6 \times 10^{-8}$): the magnitude signal is structurally sensible (`burial_analysis_2vuk.json`).
- **AlphaFold DB**: top SOD1 candidates sit at pLDDT ≈ 98.8 ; the model does not assign large effects to low-confidence regions (`alphafold_db_lookup.json`).
- **gnomAD**: TP53 is extremely loss-of-function-intolerant (pLI = 0.9996), consistent with germline depletion of core-domain destabilizing variants (`gnomad_tp53_constraint.json`).
- **STRING**: TP53/SOD1 partner landscape recorded for context (`string_network_context.json`).

8 Limitations and negative results

All of the following are preserved with quantification in `results/`: (1) v1 aromatic-identity bias (up to +1.04 kcal/mol for TRP); (2) greedy multi-mutant sums are implausible under the independence approximation (+19.5 v1, +46.0 v2); (3) p53 transfer collapses at long training (r 0.426 v1 $\rightarrow$ -0.02 v2-e300) and validation-based early stopping does not repair it; (4) proline positions are out of distribution because

the training data contain no proline mutations; (5) SOD1 $\Delta\Delta G$ prediction fails outright ($r = -0.15$, $n = 8$); (6) the documented Y220C suppressor A138G is missed; (7) scan magnitudes are inflated and only ranks should be read; (8) the apparently strong MYOGLOBIN result ($r = 0.455$) is contaminated by 41 same-protein training rows and drops to 0.24-0.29 when decontaminated.

9 External tools

Tools actually executed or queried (honest ledger: `results/EXTERNAL_TOOLS.md`): RCSB PDB (481 structures downloaded and parsed), ProtDDG-Bench, the Pucci 2018 published Table 1 (verified from PDF), FireProtDB 2.0 REST API, ThermoMutDB API, Europe PMC REST, Zenodo API, SoDCoD, Biopython (independent parser cross-check), SciPy (Spearman/Fisher statistics), scikit-learn (identity-only ridge baseline), NumPy (all model math), matplotlib (figures), pytest (hermetic tests), git/GitHub, Google Drive, pdflatex (this document). Staged and planned entries are listed separately in the ledger and are not counted here.

Appendix A: External tools actually used (n=29)

Each row is an executed use; evidence files are cited in `results/EXTERNAL_TOOLS.md` and the individual `results/*.json` files. Tools listed as planned/staged in the ledger but not executed are NOT counted.

#	Tool	What it was used for
1	RCSB PDB / data.rcsb.org	481 WT+mutant structures downloaded, parsed (graph construction)
2	ProtDDG-Bench (github.com/protddg-bench)	S2648/Ssym/P53 train-test protocol
3	Pucci et al. 2018 Bioinformatics (bty348) published Table 1	benchmark reference values (PoPMuSiCsym 0.48/1.62 etc.), verified from PDF
4	FireProtDB 2.0 REST API	SOD1 mutation search + mutant detail endpoints
5	ThermoMutDB API	full dump, human SOD1 (P00441) ddG extraction
6	Europe PMC REST API	article lookup (PMID/PMC resolution)
7	Zenodo API	ThermoMPNN FireProtDB+PDB dataset (record 8169289)
8	NumPy	all model math (GCN layers, Adam, backprop)
9	pytest	24 hermetic unit tests
10	git + GitHub	versioned delivery (SSH, ed25519)
11	Google Drive	results delivery folder (bundle uploads)
12	SoDCoD (fujisawagroup)	SOD1 mutant conformation metadata (lookup)
13	FireProtDB OpenAPI/Swagger spec	API query construction
14	matplotlib	paper figures (fig1 scatter, fig2 bias bars, fig3 curves)
15	pdflatex / TeX Live (mathptmx)	paper PDF compilation
16	Biopython (Bio.PDB)	independent cross-check of our PDB parser; exposed the altloc-CA duplicate bug (2VUK 197 vs 195 residues); confirms 195-residue chain + superstable background in stats_and_baseline
17	SciPy (scipy.stats)	Spearman rho/p-value, Fisher-z CI on benchmark r
18	scikit-learn (Ridge)	identity-only no-structure baseline on same train/test rows (r_inv 0.251 vs GNN 0.592)
19	UniProt REST API	P04637/P00441 feature cross-check: all 11 p53 scan positions carry curated cancer-variant annotations; R273/Y220 are LFS germline hotspots; SOD1 H46R/G85R/V148I ALS1-annotated (res
20	PDBe API (SIFTS)	PDB<->UniProt numbering verification: 2VUK and 1SPD author numbering = UniProt canonical at all 10 checked positions (results/pdbe_sifts_check.json); guards the SOD1 numbering-ambi
21	FreeSASA 2.2.1	per-residue SASA on 2VUK; burial correlates with predicted lddG (Spearman 0.391, p=1.6e-8) (results/burial_analysis_2vuk.json)
22	pandas	scan/candidate table joins and exports (results/scan_candidates_2vuk_top250.csv, burial analysis)

#	Tool	What it was used for
23	AlphaFold Protein Structure DB	AF-P00441-F1 model downloaded; pLDDT-vs-scan analysis: top SOD1 candidates sit in high-confidence regions (pLDDT 98.8); pLDDT/lddG Spearman 0.295 (results/alphafold_db_lookup.json)
24	Pfam via EBI InterPro API	domain membership: all 11 p53 scan positions inside PF00870 (P53 DNA-binding); SOD1 positions inside PF00080 except A4/C6 (N-terminal, outside domain - coincides with the two worst)
25	RCSB PDB Data API (entry annotations)	experimental metadata: 2VUK 1.5A Y220C+stabilizing-drug complex, 1SPD 2.4A, 3ECU apo 1.9A, 1N18 C6A/C111S 2.0A (results/rcsb_entry_metadata.json)
26	NCBI E-utilities (PubMed eSearch/ESummary)	literature verification for A4V apo-stability caveat (3 hits, titles committed); p53 suppressor phrase queries returned 0 hits (recorded honestly) (results/ncbi_eutils_litcheck.json)
27	gnomAD GraphQL API	TP53 germline constraint: pLI 0.9996, mis_z 1.12, oe_lof 0.258 (results/gnomad_tp53_constraint.json)
28	PyMOL open-source 3.x	structure figure fig4 (2VUK candidate geometry, paper/figs/fig4_candidates_2vuk.png; session build/pymol_2vuk_session.pse)
29	STRING DB API v12	TP53/SOD1 functional-network context (results/string_network_context.json)

Appendix B: Per-identity bias, v1 vs v2 (S2648)

Mean residual (predicted – experimental $\Delta\Delta G$, kcal/mol) by mutant amino-acid identity. The v1 model systematically over-credits aromatic/hydrophobic identities; the v2 center-aware readout suppresses this by an order of magnitude.

Mutant identity	<i>n</i> (S2648)	v1 mean resid	v2 mean resid
ALA	744	+0.195	-0.160
ARG	51	+0.459	+0.127
ASN	73	+0.354	+0.062
ASP	74	+0.327	-0.022
CYS	66	+0.604	-0.072
GLN	89	+0.086	+0.127
GLU	66	+0.297	-0.021
GLY	307	+0.368	-0.075
HIS	50	+0.603	-0.003
ILE	118	+1.001	+0.191
LEU	155	+0.656	-0.006
LYS	70	+0.257	+0.016
MET	51	+0.501	-0.070
PHE	182	+0.602	-0.076
SER	129	+0.535	-0.005
THR	106	+0.805	-0.029
TRP	40	+1.038	-0.061

Mutant identity	<i>n</i> (S2648)	v1 mean resid	v2 mean resid
TYR	52	+0.882	+0.042
VAL	221	+0.877	+0.073

Appendix C: Top novel 2VUK rescue candidates (bias-corrected, PRO-masked)

Full ranking in build/scan_2vuk_whole_v2.json; top 250 in results/scan_candidates_2vuk_top250.csv. Ranks, not magnitudes, are the output.

Rank	Mutation	$\Delta\Delta G_{\text{pred}}$	$\Delta\Delta G_{\text{corr}}$
1	SER240CYS	+6.452	+6.524
2	GLY245CYS	+5.981	+6.052
3	MET246CYS	+5.947	+6.019
4	ARG273CYS	+5.842	+5.913
5	SER241CYS	+5.527	+5.599
6	ALA189ARG	+5.608	+5.481
7	MET243CYS	+5.153	+5.224
8	ALA161GLU	+5.148	+5.169
9	ALA161TYR	+5.189	+5.147
10	ASN247CYS	+5.039	+5.110
11	ALA138GLU	+5.061	+5.083
12	ALA138CYS	+4.973	+5.045
13	ALA161ARG	+5.059	+4.931
14	THR123CYS	+4.806	+4.878
15	TYR239CYS	+4.346	+4.418
16	ARG110ILE	+4.509	+4.318
17	GLY244CYS	+4.244	+4.316
18	ALA189TYR	+4.332	+4.290
19	SER215LEU	+4.231	+4.237
20	GLY279MET	+4.153	+4.223
21	ALA161HIS	+4.197	+4.200
22	ALA129HIS	+4.172	+4.175
23	ALA276CYS	+4.103	+4.175
24	HIS178CYS	+4.080	+4.152
25	ALA138MET	+4.077	+4.147

Appendix E: Top WT-SOD1 (1SPD) stabilizing candidates

Hypotheses only - the model does not validate on SOD1 (Sec. 5.6); listed so future work can test or refute them.

Rank	Mutation	$\Delta\Delta G_{\text{pred}}$	$\Delta\Delta G_{\text{corr}}$
1	GLY82ASN	+5.878	+5.816
2	GLY33TYR	+5.738	+5.696
3	GLY33TRP	+5.277	+5.338
4	GLY82LYS	+5.157	+5.141
5	GLY41TRP	+5.001	+5.062

Rank	Mutation	$\Delta\Delta G_{\text{pred}}$	$\Delta\Delta G_{\text{corr}}$
6	GLY82HIS	+4.932	+4.935
7	GLY41ILE	+5.044	+4.853
8	GLY82LEU	+4.824	+4.830
9	ALA4TYR	+4.866	+4.824
10	GLY82ARG	+4.756	+4.629
11	GLY41PHE	+4.385	+4.460
12	ALA4TRP	+4.314	+4.375
13	GLY147CYS	+4.223	+4.295
14	ASN19TYR	+4.214	+4.172
15	ALA145TYR	+4.186	+4.144
16	GLY82THR	+4.109	+4.139
17	ALA145CYS	+4.058	+4.130
18	ALA4LEU	+4.098	+4.103
19	ALA4CYS	+3.976	+4.048
20	GLY127LEU	+4.007	+4.013
21	GLY82ILE	+4.176	+3.986
22	ALA60ARG	+4.112	+3.985
23	GLY33PHE	+3.905	+3.981
24	GLY16CYS	+3.904	+3.976
25	THR137TRP	+3.868	+3.929

Appendix F: Reproducibility

Environment

Python 3.10, NumPy, SciPy, scikit-learn, pandas, Biopython, FreeSASA 2.2.1, PyMOL open-source, matplotlib, TeX Live (mathptmx). CPU-only; no GPU. Sandbox: 2 CPU / 2 GB RAM.

Exact commands

- Training (benchmark model): `python3 scripts/train_bench.py -arch v2 -epochs 300 -hidden 64 -mlp-hidden 32 -lr 0.001 -batch 32 -seed 0 -ckpt build/ckpt_v2.pkl -out build/bench_v2_e`
- v1 model: same with `-arch v1 -epochs 150 -hidden 32 -mlp-hidden 16 -lr 0.0005`.
- Validation-split study: `-val-frac 0.15 -val-every 5` (PDB-grouped split, seed 0).
- Bias: `scripts/bias_measure.py`; scans: `scripts/rescue_scan_v2_whole.py` (2VUK, 1SPD); validation: `scripts/sod1_validate.py`; statistics: `scripts/stats_and_baselines.py` (5000-resample bootstrap, seed fixed); figures: `scripts/make_figures.py`.
- Tool CLI: `tools/gnn-ddg scan -pdb 2VUK -chain A`.
- Tests: `python3 -m pytest tests/` (27 hermetic tests, no network).

Leakage protocol

S2648/Ssym share no exact (structure, position, WT, mutant) rows on the inverse split (verified: 0/342 overlap, committed in `results/ssym_benchmark_v2_e300.json`). Direct-split overlap with published

training sets is expected and is why the inverse split is the reported benchmark. Checkpoints, predictions and every quoted number regenerate deterministically from the committed scripts with the seeds above; the altloc parser bug and its superseded artifacts are preserved in build/superseded_altloc/ as an audit trail.

Appendix D: Accession-level datasets used (n=493)

Counted at accession level per the program rule: each PDB structure, benchmark subset, database dump and per-protein resource counts as one dataset. Provenance per entry: results/DATASETS.md.

ProtDDG-Bench S2648 (training, PMID 21569468) - 2648 mutations	ProtDDG-Bench Ssym direct (342 rows)
Pucci et al. 2018 Table 1 published benchmark values (transcribed from PDF)	ThermoMutDB human SOD1 subset (23 rows)
SoDCoD SOD1 conformational database (188 mutants, metadata)	UniProt P00441 (SOD1 sequence/reference)
107L	108L
111L	112L
115L	118L
122L	123L
127L	128L
150L	172L
1AEP	1AG2
1AM7	1AMQ
1APS	1ARR
1B7S	1B8E
1BLC	1BNI
1BPT	1BRG
1BRJ	1BRK
1BSC	1BSD
1BTI	1BVC
1CEY	1CHK
1CTW	1CU0
1CU5	1CU6
1CUQ	1CV0
1CV4	1CV5
1D2W	1D2Y
1DMM	1DMQ
1DYC	1DYG
1DYG	1E65
1E6M	1EY0
1EY6	1EY7
1FLQ	1FLU
1FN5	1FNA
1G06	1G07
1G0K	1G0L
1G0Q	1G1V
1G6N	1GAY
1GB3	1GB5
1GB8	1GB9

1GBX	1GBY
1GF9	1GFA
1GFH	1GFJ
1GFT	1GFU
1H7M	1HEM
1HFZ	1HIB
1HMK	1HMS
1HUU	1I6S
1IGV	1IHB
1IO2	1IOB
1IR7	1IR8
1JIW	1K9Q
1KDX	1KE4
1KVB	1KVC
1L03	1L04
1L08	1L09
1L12	1L13
1L16	1L17
1L20	1L21
1L24	1L33
1L39	1L42
1L46	1L47
1L53	1L55
1L59	1L60
1L63	1L65
1L68	1L69
1L85	1L86
1L90	1L91
1L94	1L95
1L99	1LAV
1LHH	1LHI
1LHL	1LNI
1LVE	1LYE
1LYH	1LYI
1MBG	1MGR
1MX2	1MX4
1NAG	1OH0
1OTR	1OUA
1OUD	1OUE
1OUH	1OUI
1P2P	1P2R
1P64	1P6Y
1PGA	1POH
1QIS	1QIT
1QND	1QS5
1QT6	1QT7
1QTD	1RBR

1RBU	1RBV
1RDC	1RG8
1RIS	1RN1
1RTP	1SAK
1SPD	1SSO
1SYE	1SYG
1TLA	1TPK
1TYV	1UBQ
1VQG	1VQH
1WIT	1WQ5
1WQO	1WQP
1YAM	1YAN
1YAQ	1YHB
1ZG4	1ZNJ
221L	224L
232L	233L
236L	237L
240L	241L
244L	245L
253L	254L
2A36	2ABD
2DRI	2EXZ
2EY5	2EY6
2EYJ	2EYL
2EYP	2F0D
2F0G	2F0H
2H61	2HEA
2HED	2HEE
2IMM	2J1X
2NVH	2OCJ
2TRT	2TRX
2XWR	3AA2
3AA5	3C7W
3C81	3C83
3C8S	3CDO
3CDT	3CDV
3FI5	3GLY
3PGK	3SIL
5CRO	5DFR
8PTI	

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